Tutorial-8

Question 1

Given

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \qquad P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Find PA. Also state the order in which the rows of A have been permuted.

Solution:

Left multiplication by P rearranges the rows of A. Direct multiplication gives

$$PA = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{bmatrix}.$$

Hence the row order is

$$(R_1, R_2, R_3) \longrightarrow (R_2, R_3, R_1).$$

Question 2

Using the Gauss-Jordan elimination method, find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 5 & 4 \end{bmatrix}.$$

Solution:

To find A^{-1}, augment A with the identity matrix and reduce the left side to I:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 0 \\ 3 & 5 & 4 & 0 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 3R_1 :$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & -1 & 1 & -3 & 0 & 1 \end{array}\right].$$

Next,

$$R_3 \leftarrow R_3 - R_2,$$

so

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right].$$

Now use

$$R_2 \leftarrow -R_2,$$

which gives

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right].$$

Finally,

$$R_1 \leftarrow R_1 - 2R_2, \qquad R_1 \leftarrow R_1 - R_3,$$

so that

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 3 & -1 \\ 0 & 1 & 0 & 2 & -1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right].$$

Therefore,

$$\boxed{A^{-1} = \begin{bmatrix} -2 & 3 & -1 \\ 2 & -1 & 0 \\ -1 & -1 & 1 \end{bmatrix}}$$

Question 3

Determine whether the following matrix is invertible:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix}.$$

If the inverse does not exist, give a suitable reason.

Solution:

The second row is twice the first row:

$$R_2 = 2R_1.$$

Thus the rows are linearly dependent. Equivalently, row reduction gives

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 3 & 4 \end{bmatrix}.$$

Since a zero row appears, rank$(A) < 3$, so A is singular and therefore

$$\boxed{A^{-1} \text{ does not exist.}}$$

Question 4

Given

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix},$$

find A^{-1} and A^T. Hence, verify that

$$(A^{-1})^T = (A^T)^{-1}.$$

Solution:

Finding A^{-1} using Gauss-Jordan elimination only:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 3R_1 :$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right].$$

Then

$$R_2 \leftarrow -R_2,$$

which gives

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

Finally,

$$R_1 \leftarrow R_1 - 2R_2 :$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

Hence

$$\boxed{A^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}}.$$

Also,

$$\boxed{A^T = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}}.$$

Therefore,

$$(A^{-1})^T = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}.$$

To obtain $(A^T)^{-1}$, again use Gauss-Jordan elimination:

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 5 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right].$$

Using

$$R_2 \leftarrow -R_2, \qquad R_1 \leftarrow R_1 - 3R_2,$$

we obtain

$$\left[\begin{array}{cc|cc} 1 & 0 & -5 & 3 \\ 0 & 1 & 2 & -1 \end{array}\right].$$

Thus

$$(A^T)^{-1} = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix} = (A^{-1})^T.$$

Hence,

$$\boxed{(A^{-1})^T = (A^T)^{-1}}.$$

Question 5

Consider a point $P = (2, 3)$.

(a) Find the transformation matrix for a rotation of $90°$ anticlockwise about the origin.

(b) Find the transformation matrix for a horizontal shear with shear factor 2.

(c) Find the transformation matrix for scaling the x-coordinate by 3 and the y-coordinate by 2.

Solution:

Using the column-vector convention $\begin{bmatrix} x' & y' \end{bmatrix}^T = T \begin{bmatrix} x & y \end{bmatrix}^T$:

(a) Rotation by $90°$ anticlockwise

$$R(90°) = \begin{bmatrix} \cos 90° & -\sin 90° \\ \sin 90° & \cos 90° \end{bmatrix} = \boxed{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}.$$

For $P = (2, 3)$,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}.$$

(b) Horizontal shear with shear factor 2

For a horizontal shear, $x' = x + 2y$ and $y' = y$. Therefore,

$$\boxed{H = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}}.$$

For $P = (2, 3)$,

$$H \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}.$$

(c) Scaling by 3 in x and 2 in y

$$\boxed{S = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}}.$$

For $P = (2, 3)$,

$$S \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}.$$

Question 6

Consider the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}.$$

Determine whether A^{-1} exists. Justify your answer using the following methods:

1. Determinant
2. Rank
3. Linear dependence of rows/columns

4. Gauss-Jordan elimination
5. Singular matrix definition

Solution:

The inverse does not exist. Each requested method shows that A is singular.

1. Determinant

$$\det(A) = \begin{vmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{vmatrix}.$$

Expanding along the first row,

$$\begin{aligned} \det(A) &= 2\begin{vmatrix} 2 & 3 \\ 6 & 9 \end{vmatrix} - 4\begin{vmatrix} 1 & 3 \\ 3 & 9 \end{vmatrix} + 6\begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} \\ &= 2(18-18) - 4(9-9) + 6(6-6) = 0. \end{aligned}$$

Hence

$$\boxed{\det(A) = 0},$$

so A^{-1} does not exist.

2. Rank

Row reducing,

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence

$$\boxed{\operatorname{rank}(A) = 1 < 3},$$

so A is not invertible.

3. Linear dependence of rows/columns

The rows satisfy

$$R_1 = 2R_2, \qquad R_3 = 3R_2.$$

The columns also satisfy

$$C_2 = 2C_1, \qquad C_3 = 3C_1.$$

Thus the rows and columns are linearly dependent, so A is not invertible.

4. Gauss-Jordan elimination

Augment A with I:

$$\left[\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 3 & 6 & 9 & 0 & 0 & 1 \end{array}\right].$$

First swap R_1 and R_2:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 0 & 1 & 0 \\ 2 & 4 & 6 & 1 & 0 & 0 \\ 3 & 6 & 9 & 0 & 0 & 1 \end{array}\right].$$

Then apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 3R_1 :$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & -3 & 1 \end{array}\right].$$

The left side cannot be reduced to I_3. Hence the Gauss-Jordan process cannot produce $[I_3 \mid A^{-1}]$, and therefore A^{-1} does not exist.

5. Singular matrix definition

A square matrix is singular if its determinant is zero, equivalently if it does not possess an inverse. Since

$$\det(A) = 0,$$

A is singular. Therefore,

$$\boxed{A^{-1} \text{ does not exist.}}$$

Question 7

Let A be the matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}.$$

(a) Using the Gauss-Jordan elimination method, find the inverse of matrix A.

(b) Find the transformation matrix for a horizontal shear with shear factor 3.

Solution:

(a) Inverse using Gauss-Jordan elimination

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 5 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1 :$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

Then

$$R_1 \leftarrow R_1 - 2R_2 :$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 5 & -2 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

Therefore,

$$\boxed{A^{-1} = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}}.$$

(b) Horizontal shear with shear factor 3

For a horizontal shear, $x' = x + 3y$ and $y' = y$. Hence the transformation matrix is

$$\boxed{H = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}}.$$